

BASEL-BINNEN, Switzerland

WMO: 066010

Lat: 47.5411N

Lon: 7.5836E

Elev: 1040

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.8	(c) 22.4	(d) 11.1	(e) 10.1	(f) 22.6	(g) 15.5	(h) 12.5	(i) 25.8	(j) 24.3	(k) 52.2	(l) 20.8	(m) 48.1	(n) 5.0	(o) 100	(p) 0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	89.1	69.1	85.3	67.9	82.0	66.7	70.7	84.5	69.1	82.1	67.7	79.7	4.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	100.0	75.8	64.7	95.0	73.8	63.5	91.0	72.7	35.1	85.2	33.8	82.2	32.6	80.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.4	(b) 14.3	(c) 11.8	(e) 13.7	(f) 94.8	(g) 6.4	(h) 3.0	(i) 9.1	(j) 97.0	(k) 5.3	(l) 98.7	(m) 1.7	(n) 100.4	(o) -3.0	(p) 102.7		
(a) 18.4	(b) 14.3	(c) 11.8	(e) 12.8	(f) 73.9	(g) 6.5	(h) 1.4	(i) 8.1	(j) 74.9	(k) 4.3	(l) 75.7	(m) 0.7	(n) 76.4	(o) -4.0	(p) 77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	36.1	37.8	44.8	52.3	58.5	66.1	69.0	68.0	60.7	53.4	43.8	37.5
	DBStd	13.55	7.84	7.92	7.27	6.86	6.55	6.62	5.85	6.02	5.91	6.71	6.58	7.18
	HDD50	1674	433	345	190	55	7	0	0	0	2	46	205	391
	HDD65	5016	895	763	626	382	219	66	25	36	153	361	637	853
	CDD50	2555	3	2	29	124	271	482	590	559	324	151	18	2
	CDD65	424	0	0	0	1	18	98	150	130	26	1	0	0
	CDH74	3632	0	0	0	36	198	849	1328	1043	169	9	0	0
	CDH80	1156	0	0	0	1	34	269	464	365	23	0	0	0
	WSAvg	4.8	5.8	5.9	5.6	4.7	4.4	4.4	4.2	3.9	3.6	4.1	4.9	5.9
	PrecAvg	30.5	1.9	2.0	2.0	2.4	3.1	3.5	2.9	4.1	2.1	1.9	2.8	1.8
(15) Precipitation	PrecMax	42.1	3.7	6.0	4.0	4.6	5.1	7.2	5.3	7.1	6.0	4.2	7.0	3.3
	PrecMin	20.4	0.2	0.4	0.4	0.6	1.6	0.4	1.7	1.1	0.4	0.5	0.4	0.2
	PrecStd	5.2	0.9	1.3	0.9	1.1	1.0	1.5	0.9	1.7	1.3	1.0	1.6	0.8
	PrecStd	5.2	0.9	1.3	0.9	1.1	1.0	1.5	0.9	1.7	1.3	1.0	1.6	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	59.8	69.3	78.4	84.7	91.7	93.9	94.2	83.1	74.5	63.4	57.4
		MCWB	50.7	48.1	53.4	58.8	65.8	70.3	70.3	69.5	65.6	61.1	54.3	50.1
	2%	DB	54.2	55.8	64.7	73.7	79.1	86.9	89.2	88.2	79.1	69.6	59.3	53.6
		MCWB	48.2	47.0	50.6	56.3	62.4	68.7	68.5	69.4	65.0	59.5	52.6	48.7
	5%	DB	50.6	52.5	60.6	69.4	75.4	82.9	85.1	83.8	75.1	66.1	55.7	50.3
		MCWB	45.6	45.5	49.7	55.2	60.5	67.5	67.6	67.8	63.4	58.4	50.8	46.1
	10%	DB	46.8	48.7	56.6	65.4	71.0	78.9	81.5	79.9	71.2	62.7	52.8	47.6
		MCWB	43.2	42.9	48.0	53.0	58.8	65.3	66.4	66.5	61.5	56.9	48.8	44.2
	0.4%	WB	51.7	49.5	54.6	60.1	67.1	72.9	72.1	72.9	68.4	63.4	55.9	51.3
		MCDB	57.1	56.2	64.3	74.8	80.9	86.4	88.2	86.9	79.3	71.5	61.2	55.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.9	47.7	52.2	57.8	64.5	70.2	70.2	70.3	66.3	61.1	53.4	49.0
		MCDB	53.6	54.8	61.0	70.4	74.7	83.4	84.3	83.9	76.2	67.0	58.0	53.0
	5%	WB	45.7	45.7	50.4	55.7	62.3	68.3	68.7	68.8	64.4	59.3	51.3	46.7
		MCDB	49.8	52.2	58.8	67.2	71.9	80.4	82.0	81.3	72.7	64.6	55.3	50.0
	10%	WB	43.4	43.4	48.6	53.9	60.2	66.5	67.3	67.2	62.6	57.4	49.2	44.3
		MCDB	46.9	48.1	56.1	63.7	68.7	76.8	79.4	78.0	69.5	62.2	52.4	47.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.8	16.0	18.2	17.4	18.0	18.4	17.6	16.5	13.6	10.2	9.0
		MCDBR	13.8	18.4	23.3	25.8	25.0	24.4	25.1	24.5	22.1	18.6	13.9	12.0
		MCWBR	9.1	11.0	12.4	12.3	11.5	10.2	9.5	9.4	10.3	10.0	8.8	8.4
	5% WB	MCDBR	12.8	17.0	20.1	23.1	21.5	21.9	21.9	22.1	19.3	16.6	12.8	11.2
		MCWBR	8.9	10.5	11.3	11.4	10.4	9.6	9.0	9.0	9.4	9.5	8.5	8.2
(40) Clear-Sky Solar Irradiance	taub		0.308	0.334	0.368	0.405	0.415	0.419	0.418	0.411	0.388	0.365	0.330	0.305
	taud		2.446	2.369	2.311	2.229	2.248	2.278	2.274	2.307	2.352	2.401	2.449	2.449
	Ebn at Noon		246	261	268	268	270	269	267	264	259	246	234	232
	Edn at Noon		21	28	35	41	42	41	41	38	33	27	21	19
	RadAvg		340	597	989	1376	1601	1885	1799	1563	1182	738	390	274
(45) All-Sky Solar Radiation	RadStd		46	96	153	206	160	173	161	149	115	69	47	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (112 neighbors)	N/A	N/A	N/A	+1.35	+0.70	+0.52	N/A	N/A	+94	+25

Nomenclature: See separate page